Structural Drivers of the U.S. Overdose Crisis: Evidence from State-Level Social Determinants

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Qiyue Yao is responsible for Part I. Introduction, Ananya Jhamb is responsible for Part II. Data, Scarlett Shi is responsible for Part III. Methodology and Part IV. Empirical Results, Yumeng Jiang is responsible for R script writing and Part V. Conclusion. We searched for data and topics together.

1. Introduction

1.1 Background

Over the past two decades, the United States has experienced a severe and escalating overdose crisis, largely driven by prescription and illicit opioids. Age-adjusted drug overdose mortality rose steadily from 1999 and surged again in the late 2010s; in 2020 alone, there were 91,799 overdose deaths, corresponding to 28.3 deaths per 100,000 people - a 31% increase from 2019 and high enough to contribute to the recent decline in U.S. life expectancy (Hedegaard et al., 2021).

A growing literature argues that overdose problems are rooted in broader social and structural conditions rather than drug supply alone. Recent work shows that overdose mortality cannot be fully explained by substance use patterns alone (Kariisa, 2022). Park et al. (2020) criticize current studies that focus narrowly on prescribing reforms and enforcement as "necessary but not sufficient," and call for shifting attention to the social determinants of health (SDOH) - "the conditions in which people are born, grow, live, work and age… shaped by the distribution of money, power and resources."

Evidence from specific domains supports this broader SDOH perspective. **Insurance coverage** shapes access to addiction treatment: adults in the U.S. who have opioid use disorder are much more likely to be uninsured or covered by Medicaid than adults without substance use disorders. Among those who say they need treatment, not ha

ving insurance is one of the main reasons they do not receive it (Becker et al., 2008). **Food insecurity**, often seen as a marker of extreme material deprivation, is independently associated with substance use problems among young adults, even after adjusting for socioeconomic status (Pryor et al., 2016). **Depression** is one of the most important predictors of prescription-opioid overdose and death. Because of this, experts recommend being especially careful when prescribing opioids to patients with mental health conditions (Agarin et al., 2015). **Sleep problems and insomnia** are both a consequence of substance use and a driver of it. They can help trigger drug misuse and make people more likely to relapse (Roehrs & Roth, 2015). Finally, **disabled** adults - who often live with chronic pain and heavy mental-health burdens - face especially high overdose risks. In Medicare, people aged 21-64 who qualify because of disability make up only about 15% of enrollees, yet they account for over 80% of opioid overdose deaths, and their age-adjusted overdose mortality is at least 4.8 times higher than in the general U.S. population (Kuo et al., 2019).

Despite this body of work, much of the evidence is either individual-level, focused on specific high-risk subgroups, or limited to cross-sectional associations. Less is known about how state-level overdose death rates co-evolve over time with broader SDOH indicators such as insurance coverage, food insecurity, depression prevalence, short sleep duration, and disability. In this project, we address this gap by asking: **How are state-year drug overdose mortality rates related to state-level measures of insurance, food security, mental health, sleep, and disability, and which of these relationships remain most robust once we account for persistent state differences and national time trends?**

1.2 Research Questions

Building on this background, we focus on two questions at the state-year level:

- Primary question

  Are drug overdose death rates systematically related to state-level socioeconomic factors (health insurance coverage, food insecurity), mental health

(depression prevalence), and physical health indicators (short sleep duration, disability status)?

- Secondary question
  Do these relationships stay stable over time, or do they change as the overdose crisis evolves?

1.3 Hypotheses

Guided by prior evidence and basic economic intuition, we state three hypotheses:

- H1 - Economic pressure.
  States with higher shares of uninsured residents and higher food insecurity will have higher overdose death rates.
- H2 - Mental and physical health.
  States with higher prevalence of depression, short sleep, and disability will have higher overdose death rates.
- H3 - Temporal trends.
  The strength of these associations may change over time; in other words, the effect of social and health factors on overdose mortality may differ across years.

1.4 Overview of Data and Methods

To answer these questions, we link two CDC data sources at the state-year level. From CDC WONDER we take annual drug overdose death counts and construct crude mortality rates per 100,000 residents. From CDC PLACES we take state-level estimates of the share of adults who are uninsured, food insecure, depressed, sleep less than seven hours, or report a disability. Our main sample covers 51 states and DC from 2011 to 2020. We treat the data as a pooled state-year panel and estimate ordinary least squares regressions with a linear time trend. Model 1 includes year and all five social determinants; Model 2 adds interactions between year and the economic variables to test whether their effects change over time.

1.5 Main Findings and Practical Implications

In the baseline model, overdose mortality is strongly and positively associated with depression, short sleep, and lack of insurance, while food insecurity shows only a weak relationship. By contrast, disability is negatively associated with overdose mortality in our state-level models, contrary to micro-level evidence and suggesting possible confounding by unobserved state characteristics. The interaction model fits the data slightly better and indicates that the association between uninsurance and overdose deaths was stronger in earlier years and has weakened over time, whereas the role of food insecurity remains small and unstable.

Taken together, these patterns suggest that state-level mental health burden and sleep disruption are closely intertwined with the overdose crisis, and that gaps in insurance coverage may matter most in the earlier phases of the epidemic. The results also imply that focusing only on opioid supply and prescribing is not enough: effective overdose prevention will likely require sustained investment in mental health services, sleep and stress management, and protection for disabled and economically vulnerable populations.

1.6 Structure of the Report

The rest of this report is organized as follows. Section 2 describes the data sources in detail. Section 3 outlines our regression models, variable construction and diagnostic checks. Section 4 presents the empirical results and robustness analysis. Section 5 discusses policy implications, limitations, and directions for future work.

2. Data

2.1 Drug Overdose Mortality Data

The data sources used in our analysis are CDC WONDER for overdose mortality and CDC PLACES for social determinants. To measure drug overdose mortality, we use data from the CDC Wide-ranging Online Data for Epidemiologic Research (CDC WONDER). It is an online database provided by the CDC that lets researchers query and download public health data, including mortality statistics.

- **CDC WONDER mortality data are derived from death certificate records collected nationwide through the National Vital Statistics System (NVSS). These include counts of deaths, population denominators by geographic area and cause of death：**[Underlying Cause of Death 1999-2020]
- This is the key dataset typically used to analyze overdose deaths. It helps select:
    - Number of deaths
    - Crude death rates per 100,000 residents
    - Age-adjusted death rates
    - Breakdown by cause of death using ICD-10 codes (for drug overdose, common codes include X40-X44, X60-X64, X85, Y10-Y14)
- Why This Source Is Suitable
    - It is nationally representative and annual, matching the 2011-2020 state-year panel.
    - It allows consistent comparisons across states and over time.

2.2 Social Determinants of Health Data

To measure social determinants, we use data from **CDC PLACES which is a public health data resource that provides modeled estimates of community health measures across the U.S. at multiple geographic levels：** [Social Determinants of Health | PLACES | CDC]

PLACES combines data from several sources to estimate local health and social measures:

- Behavioral Risk Factor Surveillance System (BRFSS)
    - A national telephone survey of adults that provides self-reported health behaviors and conditions.
    - Used with small-area estimation techniques to produce reliable estimates even in areas with limited direct survey responses.
- American Community Survey (ACS)

    - A nationally representative survey conducted by the U.S. Census Bureau.
    - PLACES uses 5-year ACS estimates to derive measures of social determinants like socioeconomic status, insurance coverage, and food security.
- We have state-level estimates for:
    - Share of adults uninsured
    - Food insecurity
    - Depression
    - Sleep < 7 hours
    - Disability
- Estimates are produced with statistical modeling (small-area estimation) combining BRFSS responses and ACS structural data so that state-level estimates reflect plausible prevalence even in areas with sparse direct survey data. cdc.gov+1

2.3 Key Features of PLACES Data

- Annual or Multi-year Estimates: Many SDOH measures are based on multi-year ACS estimates (5-year) to stabilize the data.
- Geographic Consistency: Data are available for all states and DC, enabling panel analysis.
- Modeled Estimates: PLACES uses small-area estimation techniques tailored to provide reliable estimates even where direct survey data are limited.

2.4. Integrating the Data Sources

Mortality data from CDC WONDER provide the dependent variable state-year overdose death rates per 100,000 residents. PLACES data provide independent variables social determinants measured at the state level.Both datasets span 2011-2020 and cover the 51 states + DC panel. Merging is done at the state and year level so that each observation corresponds to one state in one year with corresponding mortality and social determinant measures.

3. Methodology

3.1 Variable Definition

- Response Variable

  Death Rate of drug overdose. The crude rate is calculated as the number of drug overdose deaths out of 100,000 population of a given state in the same year. This measure represents the percentage of morality and effectively eliminates statistical bias arising from differences in population size across states.

- Core Explanatory Variable:
  - Economic Pressure in certain region

    Regional economic pressure is captured by indicators related to health insurance coverage and food security. Specifically, the level of economic stress faced by adults in each state is measured by the percentage of individuals lacking health insurance and the percentage experiencing difficulties in purchasing basic food necessities.
  - Individual Health Status

    Individual health status is proxied by indicators that are strongly associated with psychological well-being, including depression prevalence, average sleep duration among adults, and disability status. These variables collectively reflect the mental and physical conditions closely linked to mental health outcomes.

Table 3.1: Key Variables Explanation

Crude_Rate	Measures the regional drug overdose death rate. Calculated as the proportion of drug overdose deaths relative to 100,000 population
Lack_Insurance	An indicator of economic pressure. Lack of health insurance reflects greater financial stress on individuals. This variable is calculated as percentage of adults age 18-64 without health insurance.
Food_Insecurity	An indicator of economic pressure. Individuals who have faced difficulty in purchasing essential food for survival within the past 12 months provide a strong representation of personal economic burden. Calculated as the percentage of adults experiencing fo

	od insecurity in the past 12 months.
Depression	An indicator of mental health status. The prevalence of depression directly reflects poor mental health conditions. Calculated as the percentage of individuals estimated "ever told they had a depressive disorder".
Short sleep	Sleep duration is closely related to mental health status. Insufficient sleep has adverse effects on psychological well-being. Therefore, whether adults obtain at least seven hours of sleep is used as one indicator of mental health. Calculated as the percentage of adults with short sleep duration (less than 7 hours)
Disability	Disability status is also associated with mental health conditions. We use **Disability**, which integrates cognitive disability and other disability in the original dataset. Calculated as the percentage of individuals with disabilities
Year	Considering that regional economic conditions can evolve over time, the observations might inherently exhibit time trends. Therefore, we introduce a time variable to capture these temporal dynamics.

3.2 Model Setup

Based on our empirical analysis, we construct the following baseline regression ***Model I***:

$$\text{Crude_Rate} = \beta 0 + \beta 1*\text{Year} + \beta 2*\text{Lack_Insurance} + \beta 3*\text{Food_Insecurity} + \beta 4*\text{Depression} + \beta 5*\text{Short sleep} + \beta 6*\text{Disability} + \varepsilon$$

Meanwhile, to examine whether the impact of regional economic factors on drug overdose death rates changes over time, we introduce interaction terms between regional economic factors and time (Year) and construct ***Model II***:

$$\text{Crude_Rate} = \beta 0 + \beta 1*\text{Year*Lack_Insurance} + \beta 2*\text{Year*Food_Insecurity} + \beta 3*\text{Depression} + \beta 4*\text{Short sleep} + \beta 5*\text{Disability} + \beta 6*\text{Lack_Insurance} + \beta 7*\text{Food_Insecurity} + \varepsilon$$

4. Empirical results

4.1 Regression Outcome

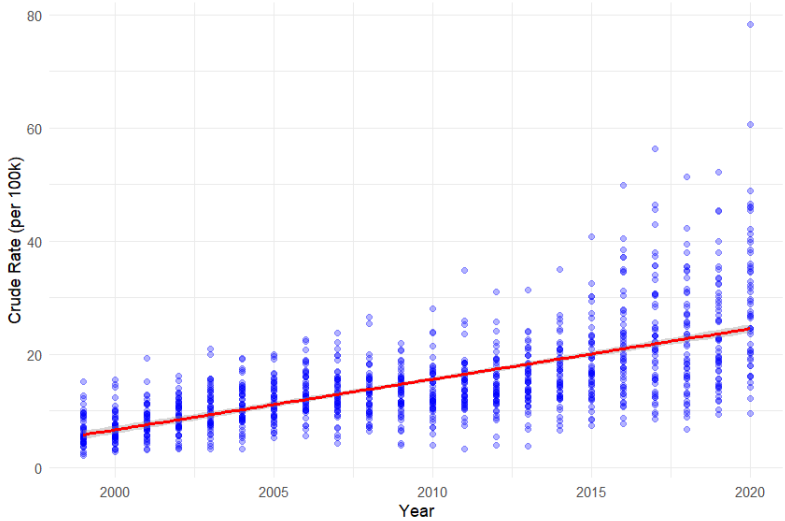
4.1.1 Descriptive Statistics

Table 4.1: Descriptive Report

	mean	min	max
Crude_rate(%)	15.16	2.00	78.30
Lack_Insurance(%)	10.40	5.15	20.70
Food_Insecurity(%)	13.78	10.28	26.42
Depression(%)	23.29	16.18	31.42
Short sleep(%)	36.05	30.44	43.02
Disability(%)	32.12	21.80	40.58

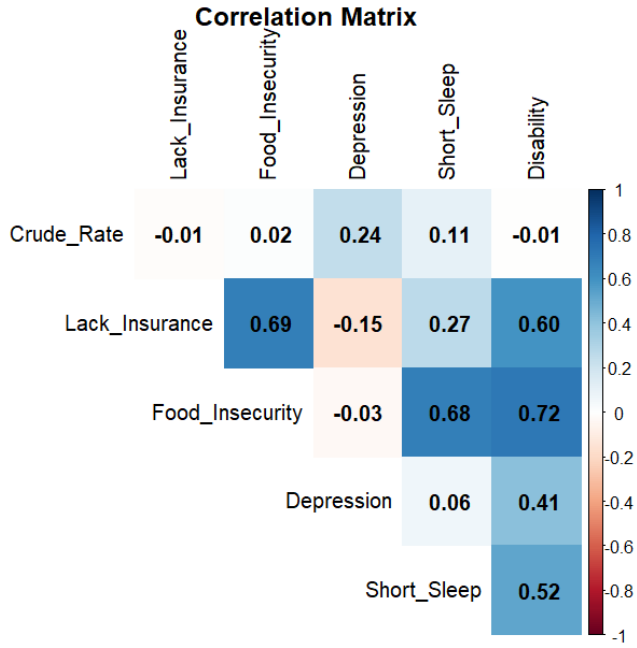
Also, we found that there is an increasing trend for Drug overdose death rate from 1999-2020:

Table 4.2: Time trend for drug overdose death rate(1999-2020)



4.1.2 Correlation Matrix

Table 4.3: Correlation Matrix



We use dark blue to represent strong positive correlation and dark red to represen t strong negative correlation. According to the correlation matrix, crude rate has most strong correlation with depression(0.24)

4.1.3 Regression outcome for Model I:

Table 4.4

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-1886.1043	61.1739	-30.8318	1.46E-141***
Year	0.9267	0.0304	30.4987	1.9782E-139***
Lack_Insurance	1.0020	0.1262	7.9373	6.36638E-15***
Food_Insecurity	0.1778	0.1011	1.7596	0.078836296
Depression	1.4549	0.0876	16.6085	5.32403E-54***
Short_Sleep	0.7416	0.0963	7.6983	3.75157E-14***
Disability	-1.0694	0.0832	-12.8588	9.1665E-35***
Multiple R-squared:0.5877			Adjusted R-squared: 0.5848	

F-statistic: 206.2 on 6 and 868 DF, p-value: < 2.2e-16

Table 4.3 reports the core regression results examining the relationship between drug overdose death rates and residents' economic pressure and mental health status. The p-value of this whole model is 2.2e-16, which means our model is significant and there is a linear relationship between our response variable and core explanatory variable. And 58.48% of the validation in drug overdose death rate can be explained by explanatory variables above.

The coefficient for Year is 0.92 and it pass the 1% level of significance, implying an average natural increase in drug overdose death rates of about 0.92 units per year, which is the strongest driver.

In the baseline regression, the coefficient on **Depression** is 1.455, the coefficient on **Lack_Insurance** is 1.002, and the coefficient on **Short_Sleep** is 0.741; all these predictors pass the 1% level of significance. This indicates that higher regional depression prevalence, greater economic pressure among residents are associated with a significant increase in the drug overdose death rate. After controlling for time trends, for every 1% increase in the depression rate, the death rate increases by an average of 0.46 units, with high statistical significance. These variables exhibit a strong positive correlation with overdose deaths. Consequently, it provides strong support for **Hypothesis H1**.

4.1.4 Regression outcome for Model II

Table 4.5

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-3139.4449	266.3773	-11.7857	7.6773E-30***
Year	1.5503	0.1325	11.6976	1.88523E-29***
Lack_Insurance	161.9079	32.1327	5.0387	5.70551E-07***

Food_Insecurity	-21.7552	20.3557	-1.0688	0.285479642
Depression	1.4586	0.0860	16.9694	5.61503E-56***
Short_Sleep	0.7435	0.0945	7.8667	1.08257E-14***
Disability	-1.0727	0.0816	-13.1446	4.0786E-36***
Year:Lack_Insurance	-0.0801	0.0160	-5.0075	6.6826E-07***
Year:Food_Insecurity	0.0109	0.0101	1.0776	0.281520712

Multiple R-squared:0.604 Adjusted R-squared: 0.6003

F-statistic: 165.1 on 8 and 866 DF, p-value: < 2.2e-16

Table 4.4 reports the core regression results examining the relationship between drug overdose death rates and residents' economic pressure, mental health status and interactions of time and economic pressure.¥ The p-value of this whole model is 2.2e-16, which means our model is significant and there is a linear relationship between our response variable and core explanatory variable. After introducing the interaction terms, the model's R-squared increases to 0.60. And 60.03% of the validation in drug overdose death rate can be explained by explanatory variables above.

The interaction term between **Year** and **Lack_Insurance** **has a coefficient of −0.08 and is statistically significant at the 1% level. This interaction effect provides strong evidence that the impact of lacking health insurance on drug overdose death rate is not constant over time but evolves dynamically. Specifically, the negative coefficient indicates that while lack of health insurance remains a risk factor, its marginal effect on overdose mortality diminishes over time, potentially reflecting the expansion of public health interventions, changes in insurance coverage policies, or improved access to treatment and harm-reduction programs in later years.**

4.2 Model Diagnostics and Comparison

4.2.1 Mode Comparison(ANOVA)

In addition to the Model 1, Model 2 introduced two interaction terms: Year × Lack_Insurance and Year × Food_Insecurity. The ANOVA comparison results show that Model II's residual sum of squares (RSS) decreased from 28132 to 27020, with a reduction of 2 degrees of freedom. The sum of squares added was 1111.8, corresponding to an F-statistic of 17.817. The model's explanatory power was enhanced since R-square increased after the inclusion of the interaction terms.

4.2.2 Multicollinearity Check (VIF)

Year	Lack_Insurance	Food_Insecurity	Depression	Short_Sleep	Disability
1.0002	2.8277	4.5203	1.9927	2.2492	4.2742

We performed a VIF check on the independent variables in the regression Model 1. Since the Variance Inflation Factor values for all variables are all less than 5, Model 1 does not suffer from severe multicollinearity issues. This implies that the estimates of the regression coefficients are relatively stable.

5. Conclusion and Policy Implications

5.1 Conclusion

This study investigated the structural determinants of the U.S. drug overdose crisis by integrating state-level mortality data with socioeconomic and health indicators from 2011 to 2020. By constructing multivariate regression models with temporal interactions, we derived three critical conclusions:

First, **mental and behavioral health are key drivers.** Our analysis confirms that Depression and Short Sleep Duration are the most robust predictors of overdose mortality. This validates that the crisis is not merely about substance abuse, but a symptom of broader psychological distress and lifestyle deterioration.

Second, **the protective role of health insurance has weakened over time.** While lacking health insurance is a risk factor, our interaction model shows its impact has significantly diminished in recent years. This suggests that while expanding insurance coverage was critical in the early stages of the epidemic, it is no longer a "silver bullet." As the crisis evolves, simply having insurance is not enough to protect vulnerable populations without improving the quality of care.

Third, **state-level patterns do not equal individual risks.** We found a surprising negative link between disability rates and overdose deaths at the state level, which contradicts the known high risk for disabled individuals. This highlights a critical lesson in data analytics: we cannot simply assume that trends seen in whole states apply directly to individuals. Factors like an aging population in high-disability states may be masking the true risks, warning policymakers to look beyond simple averages.

5.2 Policy Implications

Based on these findings, we propose a shift from "Substance-Focused" to "**Whole-Health**" intervention strategies:

1. **Integrate Mental Health and Sleep Hygiene:** Since depression and sleep deprivation are significant predictors, overdose prevention funds should be allocated not just to addiction treatment centers, but also to community mental health services and public health campaigns promoting sleep hygiene. Treating the underlying psychological and physiological vulnerabilities is as crucial as restricting opioid supply.
2. **Evolving Insurance Coverage to Quality of Care:** As the marginal benefit of simple insurance coverage appears to decrease over time, policy should shift focus from merely *expanding* coverage to improving the *quality* of coverage, specifically, ensuring that plans cover comprehensive mental health support, pain management alternatives, and long-term addiction recovery services.
3. **Targeted Interventions:** The strong linear time trend suggests that without structural intervention, the crisis has a natural momentum. Interventions mu

st be continuous and adapt to the changing nature of economic drivers, particularly focusing on regions with high combined risks of mental distress and economic instability.

In summary, the opioid crisis is a multi-dimensional public health challenge. Combating it requires a holistic approach that simultaneously addresses economic stability, mental well-being, and basic physiological needs like sleep.

Reference

Agarin, T., Trescot, A. M., Agarin, A., Lesanics, D., & Decastro, C. (2015). Reducing opioid analgesic deaths in America: what health providers can do. *Pain physician*, *18*(3), E307.

Becker, W. C., Fiellin, D. A., Merrill, J. O., Schulman, B., Finkelstein, R., Olsen, Y., & Busch, S. H. (2008). Opioid use disorder in the United States: insurance status and treatment access. *Drug and alcohol dependence*, *94*(1-3), 207-213.

Hedegaard, H., Miniño, A. M., Spencer, M. R., & Warner, M. (2021). Drug overdose deaths in the United States, 1999-2020.

Kariisa, M. (2022). Vital signs: drug overdose deaths, by selected sociodemographic and social determinants of health characteristics—25 states and the District of Columbia, 2019-2020. *MMWR. Morbidity and mortality weekly report*, *71*.

Kuo, Y. F., Raji, M. A., & Goodwin, J. S. (2019). Association of disability with mortality from opioid overdose among US Medicare adults. *JAMA network open*, *2*(11), e1915638-e1915638.

Park, J. N., Rouhani, S., Beletsky, L. E. O., Vincent, L., Saloner, B., & Sherman, S. G. (2020). Situating the continuum of overdose risk in the social determinants of health: a new conceptual framework. *The Milbank Quarterly*, *98*(3), 700-746.

Pryor, L., Lioret, S., van Der Waerden, J., Fombonne, É., Falissard, B., & Melchior, M. (2016). Food insecurity and mental health problems among a community sample of young adults. *Social psychiatry and psychiatric epidemiology*, *51*(8), 1073-1081.

Roehrs, T. A., & Roth, T. (2015). Sleep disturbance in substance use disorders. *The Psychiatric Clinics of North America*, *38*(4), 793.